

## Technology Stack Details

|                      |                 |                       |              |
|----------------------|-----------------|-----------------------|--------------|
| <b>Date:</b>         | 08 October 2026 | <b>Team ID:</b>       | UNM1611JASON |
| <b>Project Name:</b> | AI BlogNest API | <b>Maximum Marks:</b> | 2 Marks      |

### Technology Stack Architectural Selection:

A well-chosen technology stack ensures the application is performant, scalable, maintainable, and aligned with modern Generative AI software development practices.

#### Architecture Component / Layer Details — Part 1

| S.No | Component / Layer                                     | Technology Chosen                                                                                                                                         | Justification / Purpose                                                                                                                                                                                                                                                                                         |
|------|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <b>Frontend / Client-Side<br/>(API Consumer)</b>      | <ul style="list-style-type: none"> <li>• Thunder Client (VS Code Extension)</li> <li>• Postman Collection v2.1</li> <li>• REST HTTP Web Client</li> </ul> | <ul style="list-style-type: none"> <li>• Enables rapid automated endpoint testing and regression suites.</li> <li>• Built-in environment variable management ({{{base_url}}}, {{{token}}}).</li> <li>• Lightweight, zero-overhead testing directly from the developer workspace.</li> </ul>                     |
| 2    | <b>Backend / Server-Side<br/>(Core API Framework)</b> | <ul style="list-style-type: none"> <li>• Node.js (v20+ LTS runtime)</li> <li>• Express.js (v4.18.2)</li> <li>• MVC Architecture Pattern</li> </ul>        | <ul style="list-style-type: none"> <li>• Non-blocking, event-driven asynchronous I/O ideal for handling AI API latency.</li> <li>• Robust ecosystem, high throughput, and flexible middleware pipeline.</li> <li>• MVC pattern provides strict separation of concerns across controllers and models.</li> </ul> |

## Technology Stack Details — Part 2

### Architecture Component / Layer Details — Part 2

| S.No | Component / Layer                                        | Technology Chosen                                                                                                                                       | Justification / Purpose                                                                                                                                                                                                                                                                                            |
|------|----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3    | <b>Database / Data Storage<br/>(Persistence Layer)</b>   | <ul style="list-style-type: none"> <li>• MongoDB Atlas Cloud Cluster</li> <li>• Mongoose ODM (v7.5.0)</li> </ul>                                        | <ul style="list-style-type: none"> <li>• Flexible schema design tailored for variable-length blog content and metadata.</li> <li>• Built-in schema validation, timestamps, and relational referencing via ObjectID.</li> <li>• High availability, automated indexing, and horizontal cloud scalability.</li> </ul> |
| 4    | <b>Cloud / Hosting / Deployment<br/>(Infrastructure)</b> | <ul style="list-style-type: none"> <li>• Node.js Cloud Runtime (Render)</li> <li>• Docker Container ready</li> <li>• Localhost (Development)</li> </ul> | <ul style="list-style-type: none"> <li>• Low-latency containerized execution with auto-restart capabilities.</li> <li>• Native environment secret injection via secure cloud dashboard.</li> <li>• Zero configuration continuous deployment connected directly to Git repository.</li> </ul>                       |
| 5    | <b>Version Control &amp; CI/CD<br/>(Code Governance)</b> | <ul style="list-style-type: none"> <li>• Git &amp; GitHub</li> <li>• Semantic Versioning (v1.0.0)</li> </ul>                                            | <ul style="list-style-type: none"> <li>• Distributed version tracking, issue management, and collaborative development.</li> <li>• Clear commit traceability and clean branch management.</li> </ul>                                                                                                               |

## Technology Stack Details — Part 3 & Dependency Matrix

### Architecture Component / Layer Details — Part 3

| S.No | Component / Layer                                            | Technology Chosen                                                                                                                                                                                                | Justification / Purpose                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6    | <b>Third-Party APIs &amp; Libraries (AI &amp; Utilities)</b> | <ul style="list-style-type: none"> <li>• Google Generative AI SDK</li> <li>• jsonwebtoken (v9.0.2)</li> <li>• bcryptjs (v2.4.3)</li> <li>• express-validator (v7.0.1)</li> <li>• morgan, cors, dotenv</li> </ul> | <ul style="list-style-type: none"> <li>• <b>@google/generative-ai</b>: Direct interface to Gemini 1.5 Flash for state-of-the-art blog generation and summarization.</li> <li>• <b>jsonwebtoken &amp; bcryptjs</b>: Industry-standard cryptographic auth and token validation.</li> <li>• <b>express-validator</b>: Strict sanitization and request body validation.</li> <li>• <b>morgan &amp; cors</b>: HTTP request logging and cross-origin resource sharing.</li> </ul> |

### Package Dependency & Version Manifest (package.json):

| Package Name          | Version          | Type           | Role in AI BlogNest API                                            |
|-----------------------|------------------|----------------|--------------------------------------------------------------------|
| @google/generative-ai | ^0.24.1          | Production     | Official Google Gemini SDK for LLM generation & summarization      |
| express               | ^4.18.2          | Production     | Core web framework for HTTP routing and middleware                 |
| mongoose              | ^7.5.0           | Production     | Object Data Modeling (ODM) for MongoDB database operations         |
| jsonwebtoken          | ^9.0.2           | Production     | Generates and validates HMAC-SHA256 Bearer authorization tokens    |
| bcryptjs              | ^2.4.3           | Production     | Salted cryptographic password hashing with 10 rounds               |
| express-validator     | ^7.0.1           | Production     | Declarative request validation and error messaging middleware      |
| cors & morgan         | ^2.8.5 / ^1.10.0 | Production     | Cross-Origin Resource Sharing & development HTTP logging           |
| nodemon               | ^3.0.1           | Dev Dependency | Automatic server reload upon file modifications during development |